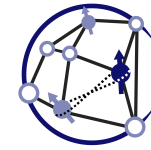


Entanglement Detection and Quantification: PT-Moments and Geometric Measure

Donghun Jung

14 Apr 2026

Department of Physics, Sungkyunkwan University
Paulee Lab, Center for Quantum Technology,
Korea Institute of Science and Technology



Paulee Group
Quantum Info with Materials and Engineering

Paper Information

Paper 1

Optimal Entanglement Certification from Moments of the Partial Transpose

Authors: Xiao-Dong Yu, Satoya Imai, and Otfried Gühne

Journal: Physical Review Letters **127**, 060504 (2021)

Paper 2

Quantifying Entanglement from the Geometric Perspective

Authors: Lisa T. Weinbrenner and Otfried Gühne

Journal: EPL **151**, 68001 (2025)

Outline

1. Review

- Separability & entanglement
- Partial transpose (PPT)
- Negativity
- Entanglement witnesses
- Multipartite entanglement

2. Paper 1: PT-Moment Certification

- PT-moments & measurement
- Classical moment problems
- Hankel matrix criteria
- Optimal p_n -OPPT criterion

3. Paper 2: Geometric Measure

- Definition & properties
- Asymptotic behaviour
- Connection to MBQC
- Tensor eigenvalues

4. Summary

Review

Entanglement Detection & Quantification

[Review] Separability & Entanglement

A bipartite state $|\psi\rangle$ is **separable** if it can be written as:

$$|\psi\rangle = \sum_i \lambda_i |\psi_A\rangle |\psi_B\rangle.$$

For density matrix, a bipartite state ρ is **separable** if it can be written as:

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B, \quad \sum_i p_i = 1, \quad p_i > 0$$

Otherwise, the state is **entangled**.

Approaches to detect entanglement

- **PNCP maps** (e.g., partial transpose): produce negative eigenvalues for entangled states
- **Entanglement witnesses**: observables with non-negative expectation on all separable states

[Review] Partial Transpose & PPT Criterion

For a bipartite state $\rho_{AB} = \sum_{ijkl} \alpha_{ijkl} |i_A\rangle\langle j_A| \otimes |k_B\rangle\langle l_B|$, the **partial transpose** on B is:

$$\rho_{AB}^{T_B} = \sum_{ijkl} \alpha_{ijkl} |i_A\rangle\langle j_A| \otimes |l_B\rangle\langle k_B|$$

which does not change eigen-spectrum for separable state.

PPT Criterion (Peres–Horodecki)

- **Separable** $\Rightarrow \rho^{T_B} \geq 0$ (positive partial transpose)
- **PPT is necessary and sufficient** for 2×2 and 2×3 systems
- In higher dimensions, PPT entangled (bound entangled) states exist

Separable vs. Entangled States

Separable States

For separable states, the partial transpose preserves positivity:

$$\begin{aligned}\rho &= \sum_i p_i \rho_i^A \otimes \rho_i^B \\ &= \sum_i p_i \left(\sum_a \lambda_i^a |\lambda_i^a\rangle \langle \lambda_i^a| \right) \otimes \left(\sum_b \mu_i^b |\mu_i^b\rangle \langle \mu_i^b| \right) \\ &= \sum_{i,a,b} p_i \lambda_i^a \mu_i^b |\lambda_i^a, \mu_i^b\rangle \langle \lambda_i^a, \mu_i^b|\end{aligned}$$

Since all coefficients $p_i \lambda_i^a \mu_i^b \geq 0$ and sum to unity, this represents a valid eigendecomposition. Under positive and trace-preserving operations, the eigenspectrum remains positive.

Separable vs. Entangled States

Entangled States

For entangled states, quasi-probability decomposition becomes necessary:

$$\rho = \sum_i q_i |a_i, b_i\rangle \langle a_i, b_i|$$

where $\sum_i q_i = 1$ but some $q_i < 0$.

The states $\{|a_i, b_i\rangle\}$ form an overcomplete (non-orthonormal) basis. Under PNCP operations, while the quasi-probability coefficients remain unchanged, the transformation of the overcomplete basis induces negative eigenvalues.

[Review] Negativity & Logarithmic Negativity

The **negativity** quantifies the violation of PPT:

$$\mathcal{N}(\rho) = \frac{\|\rho^{T_B}\| - 1}{2} = \sum_{\lambda_i < 0} |\lambda_i|$$

where $\|A\| = \text{Tr} \sqrt{A^\dagger A}$ is the trace norm.

The **logarithmic negativity** is defined as:

$$E_N(\rho_{AB}) = \ln \|\rho_{AB}^{T_B}\|$$

Properties:

- Separable states: $\|\rho^{T_B}\| = 1 \Rightarrow E_N = 0$
- Entangled states (detected by PPT): $\|\rho^{T_B}\| > 1 \Rightarrow E_N > 0$
- For the Bell state: eigenvalues of ρ^{T_B} are $\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\}$, giving $E_N = \ln 2$

[Review] Entanglement Witnesses

An observable W is an **entanglement witness** if:

- $\text{Tr}(W\rho) \geq 0$ for all separable states ρ
- $\text{Tr}(W\rho) < 0$ for some entangled state ρ

Decomposable Witnesses

A witness W is **decomposable** w.r.t. bipartition $M|\bar{M}$ if:

$$W = P_M + Q_M^{T_M}, \quad P_M \geq 0, \quad Q_M \geq 0$$

A witness is **fully decomposable** if this holds for **all** bipartitions M .

Key theorem: Fully decomposable witnesses detect **exactly** the states that are **not PPT mixtures**:

$$\rho \notin \mathcal{S}_{\text{PPT-mix}} \implies \exists W \in \mathcal{W}_{\text{fd}} : \text{Tr}(W\rho) < 0$$

[Review] Multipartite Entanglement

A state is **biseparable** if it can be written as:

$$\rho^{\text{bs}} = \sum_i p_i \rho_{M_i|\bar{M}_i}^{\text{sep}}, \quad \sum_i p_i = 1$$

A state has **genuine multipartite entanglement (GME)** iff it is NOT biseparable.

GME Detection via SDP

$$\begin{aligned} & \min_{W, P_M, Q_M} \quad \text{Tr}(W\rho) \\ & \text{subject to} \quad \text{Tr}(W) = 1 \\ & \quad \quad \quad W = P_M + Q_M^{T_M}, \quad P_M \geq 0, \quad Q_M \geq 0 \quad \forall M \end{aligned}$$

If $\min \text{Tr}(W\rho) < 0$, the state is **GME**.

Limitation: All these methods generally require **full density matrix** ρ — expensive quantum state tomography, and even numerical search for entanglement witness W .

[Review] Summary

Method	Detects	Requires	Limitation
PPT	Entanglement (NPT states)	ρ^{T_B} spectrum	Misses PPT entangled states
Negativity	+ Quantifies	ρ^{T_B} eigenvalues	Not a full measure
Witnesses	Tailored detection	Observable $\mathcal{W}$	Must construct $\mathcal{W}$ per state
SDP (GME)	Genuine multipartite	Full ρ	Exponential scaling

Question: Can we certify entanglement **without full state tomography**?

⇒ **Paper 1:** Use only the *moments* of ρ^{T_A} — accessible via randomized measurements.

Optimal Entanglement Certification from Moments of the Partial Transpose

Yu, Imai, and Gühne (PRL 2021)

Motivation: Why PT-Moments?

Challenge: Full state tomography scales exponentially. Can we detect entanglement from **partial** information?

PT-Moments

The n -th moment of the partially transposed state is:

$$p_n = \text{Tr} \left[(\rho_{AB}^{T_A})^n \right]$$

Key advantage: PT-moments can be efficiently measured via **randomized measurements** (classical shadows, U -statistics) without reconstructing ρ .

PT-Moments: Definition and Properties

Given a bipartite state ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = d$, the **PT-moments** are:

$$p_n = \text{Tr}[(\rho_{AB}^{T_A})^n], \quad n = 0, 1, 2, \dots$$

Basic properties

- $p_0 = d$ (dimension)
- $p_1 = \text{Tr}[\rho_{AB}^{T_A}] = \text{Tr}[\rho_{AB}] = 1$
- $p_2 = \text{Tr}[(\rho_{AB}^{T_A})^2] = \text{Tr}[\rho_{AB}^2]$ (purity), so $\frac{1}{d} \leq p_2 \leq 1$
- For **separable** states: $\rho_{AB}^{T_A} \geq 0$, so $p_n = \text{Tr}[(\rho_{AB}^{T_A})^n] \geq 0$ for all n , and the spectrum $(x_1, \dots, x_d)$ satisfies $x_i \geq 0$

The PT-Moment Problem

Given moments $(p_1, p_2, \dots, p_n)$, is there a **separable state** ρ_{AB} with these PT-moments?

If not $\Rightarrow$ the state **must be entangled**.

PT-Moments: Eigen-spectrum

If we are aware of all the PT-moments $p = (p_1, p_2, \dots, p_d)$ where $d = d_A d_B$, all the eigenvalues of ρ_{AB}^T can be directly calculated. It is just solving equation, employing Newton's identities.

$$p_1 = \text{Tr} \rho_{AB}^T = \sum_i \lambda_i$$

$$p_2 = \text{Tr} \rho_{AB}^{T^2} = \sum_i \lambda_i^2$$

$$p_3 = \text{Tr} \rho_{AB}^{T^3} = \sum_i \lambda_i^3$$

$$\vdots \quad \quad \quad \vdots$$

PT-Moments: Eigen-spectrum

The eigenvalues $\{\lambda_i\}$ are roots of

$$P(x) = \prod_{i=1}^d (x - \lambda_i) = x^d - e_1 x^{d-1} + e_2 x^{d-2} - \dots + (-1)^d e_d$$

where e_k are the **elementary symmetric polynomials**:

$$e_1 = \sum_i \lambda_i, \quad e_2 = \sum_{i < j} \lambda_i \lambda_j, \quad e_3 = \sum_{i < j < k} \lambda_i \lambda_j \lambda_k, \quad \dots$$

These provide a recursive relation between $s_k = \sum_i \lambda_i^k$ and e_k : $s_1 = e_1$, $s_2 = s_1 e_1 - 2e_2$, $s_3 = s_2 e_1 - s_1 e_2 + 3e_3$, $\dots$, and in general:

$$s_k = \sum_{j=1}^{k-1} (-1)^{j-1} s_{k-j} e_j + (-1)^{k-1} k e_k$$

PT-Moments: Eigen-spectrum

For example of two-qubit system,

One measure $p_1 = 1$ (trivial), $p_2 = p_2, p_3 = p_3, p_4 = p_4$. Then:

$$e_1 = 1$$

$$e_2 = \frac{p_1^2 - p_2}{2} = \frac{1 - p_2}{2}$$

$$e_3 = \frac{p_1^3 - 3p_1p_2 + 2p_3}{6} = \frac{1 - 3p_2 + 2p_3}{6}$$

$$e_4 = \frac{p_1^4 - 6p_1^2p_2 + 3p_2^2 + 8p_1p_3 - 6p_4}{24} = \frac{1 - 6p_2 + 3p_2^2 + 8p_3 - 6p_4}{24}$$

The eigenvalues of $\rho_{AB}^{T_A}$ are then the four roots of

$$x^4 - e_1x^3 + e_2x^2 - e_3x + e_4 = 0.$$

Connection to Classical Moment Problems

Writing the spectrum of $\rho_{AB}^{T_A}$ as $(x_1, x_2, \dots, x_d)$, we have:

$$p_n = \sum_{i=1}^d x_i^n$$

This connects the PT-moment problem to **classical moment problems**:

Hamburger Moment Problem: Given moments $m^{(n)} = (m_0, m_1, \dots, m_n)$, does there exist a measure on $\mathbb{R}$ with these moments?

Stieltjes Moment Problem: Same question, but the measure is supported on $[0, \infty)$.

For **separable** states, $x_i \geq 0 \Rightarrow$ Stieltjes problem.

For **general** (possibly entangled) states, $x_i \in \mathbb{R} \Rightarrow$ Hamburger problem.

The key mathematical tool for solving both: **Hankel matrices**.

Why Hankel Matrices?

Gram matrix construction (Appendix A)

Use the Hilbert-Schmidt inner product $(X, Y) := \text{Tr}(X^\dagger Y)$ in operator space.

Given $\rho \geq 0$ and $X \geq 0$ satisfying $m_k = \text{Tr}(\rho X^k) = \langle \varphi | X^k | \varphi \rangle$, construct the operator sequences:

$$v = \left(\rho^{1/2}, \rho^{1/2} X, \dots, \rho^{1/2} X^{\lfloor n/2 \rfloor} \right), \quad u = \left(\rho^{1/2} X^{1/2}, \rho^{1/2} X^{3/2}, \dots, \rho^{1/2} X^{\lfloor \frac{n-1}{2} \rfloor + 1/2} \right)$$

Their Gram matrices are:

$$(v_i, v_j) = \text{Tr}(X^i \rho X^j) = \text{Tr}(\rho X^{i+j}) = m_{i+j} \quad \Rightarrow \quad [H_k]_{ij} = m_{i+j}$$

$$(u_i, u_j) = \text{Tr}(X^{i+1/2} \rho X^{j+1/2}) = m_{i+j+1} \quad \Rightarrow \quad [B_k]_{ij} = m_{i+j+1}$$

Since Gram matrices are always positive semidefinite, we get:

$$\rho \geq 0 \Rightarrow H_{\lfloor n/2 \rfloor} \geq 0, \quad \rho \geq 0 \text{ and } X \geq 0 \Rightarrow B_{\lfloor (n-1)/2 \rfloor} \geq 0$$

Hankel Matrix Criterion

Define the **Hankel matrices** from a moment sequence $m^{(n)} = (m_0, m_1, \dots, m_n)$:

$$H_k = \begin{pmatrix} m_0 & m_1 & \cdots & m_k \\ m_1 & m_2 & \cdots & m_{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ m_k & m_{k+1} & \cdots & m_{2k} \end{pmatrix}, \quad B_k = \begin{pmatrix} m_1 & m_2 & \cdots & m_{k+1} \\ m_2 & m_3 & \cdots & m_{k+2} \\ \vdots & \vdots & \ddots & \vdots \\ m_{k+1} & m_{k+2} & \cdots & m_{2k+1} \end{pmatrix}$$

Lemma (Classical moment theory)

(a) $m^{(n)} \in \mathcal{M}_n$ (Hamburger, measure on $\mathbb{R}$) $\iff H_{\lfloor n/2 \rfloor} \geq 0$ where

$$\mathcal{M}_n = \left\{ m^{(n)} \mid \text{Tr}(\sigma X^k) = m_k, \sigma \geq 0, X^\dagger = X \right\}$$

(b) $m^{(n)} \in \mathcal{M}_n^+$ (Stieltjes, measure on $[0, \infty)$) $\iff H_{\lfloor n/2 \rfloor} \geq 0$ and $B_{\lfloor (n-1)/2 \rfloor} \geq 0$

$$\mathcal{M}_n = \left\{ m^{(n)} \mid \text{Tr}(\sigma X^k) = m_k, \sigma \geq 0, X \geq 0 \right\}$$

Since separable states have $x_i \geq 0$, the Stieltjes condition applies. Violation of either Hankel condition certifies entanglement. Let $\sigma = I, X = \rho_{AB}^T$.

p_n -PPT Criterion

Theorem (Entanglement criteria from Hankel matrices)

Let $p_k = \text{Tr}[(\rho_{AB}^{T_A})^k]$ for $k = 1, 2, \dots, n$. A necessary condition for ρ_{AB} being separable is:

$$H_{\lfloor n/2 \rfloor} \geq 0 \quad \text{and} \quad B_{\lfloor (n-1)/2 \rfloor} \geq 0$$

Lowest order: p_3 -PPT criterion

For $n = 3$, the Hankel condition $H_1 \geq 0$ gives:

$$H_1 = \begin{pmatrix} p_1 & p_2 \\ p_2 & p_3 \end{pmatrix} = \begin{pmatrix} 1 & p_2 \\ p_2 & p_3 \end{pmatrix} \geq 0$$

This requires $\det(H_1) \geq 0$, yielding the p_3 -PPT criterion:

$$\boxed{p_3 \geq p_2^2}$$

If this is violated, the state is **entangled**. Higher orders ($n = 5, 7, \dots$) give strictly stronger criteria.

Optimal Solution to the PT-Moment Problem

The Hankel criteria are **not optimal** — we may assume $\sigma \neq I$.

Exact optimization formulation

For fixed $(p_1, p_2, \dots, p_{n-1})$, find the range of p_n compatible with a separable state:

$$\begin{aligned} \min_{x_i} / \max_{x_i} \quad & \hat{p}_n := \sum_{i=1}^d x_i^n \\ \text{subject to} \quad & \sum_{i=1}^d x_i^k = p_k, \quad k = 1, 2, \dots, n-1 \\ & x_i \geq 0 \quad \text{for } i = 1, 2, \dots, d \end{aligned}$$

The solutions are found **analytically** using Cramer's rule and the Vandermonde determinant. The extremal spectra have a specific structure with at most **three distinct values**.

p_3 -OPPT Criterion (Theorem 3)

Theorem (Optimal p_3 -OPPT)

(a) There exists a d -dimensional **separable (PPT)** state ρ_{AB} satisfying $p_k = \text{Tr}[(\rho_{AB}^{T_A})^k]$ for $k = 1, 2, 3$ if and only if:

$$p_1 = 1, \quad \frac{1}{d} \leq p_2 \leq 1$$

$$\alpha x^3 + (1 - \alpha x)^3 \leq p_3 \leq [1 - (d - 1)y]^3 + (d - 1)y^3$$

where:

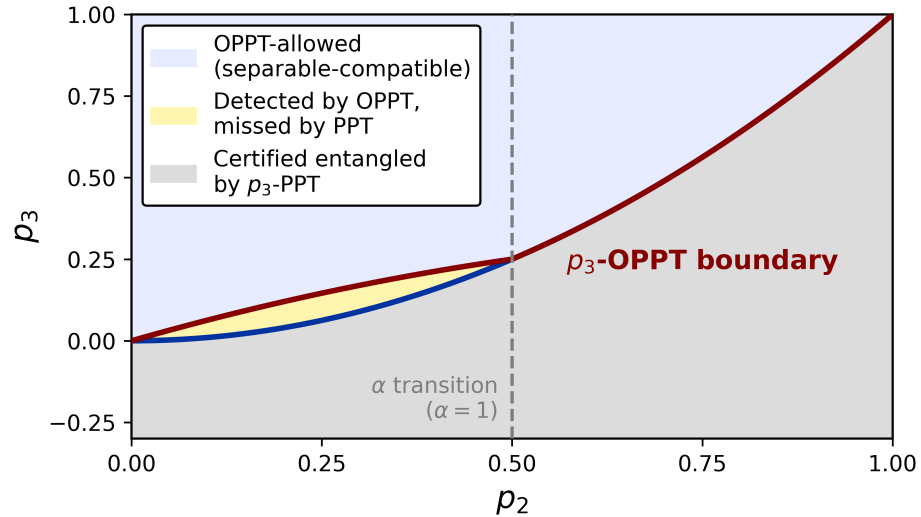
$$\alpha = \left\lfloor \frac{1}{p_2} \right\rfloor, \quad x = \frac{\alpha + \sqrt{\alpha[p_2(\alpha + 1) - 1]}}{\alpha(\alpha + 1)}, \quad y = \frac{d - 1 - \sqrt{(d - 1)(p_2 d - 1)}}{d(d - 1)}$$

(b) If the p_k for $k = 1, 2, 3$ are PT-moments from a **quantum state**, they are compatible with a separable state if and only if:

$$\boxed{p_3 \geq \alpha x^3 + (1 - \alpha x)^3}$$

The p_3 -OPPT criterion is **dimension-independent** and strictly stronger than p_3 -PPT.

Comparison: p_3 -PPT vs p_3 -OPPT



Hierarchical structure

The criteria form a hierarchy based on PT-moments:

- p_3 -PPT ($n = 3$, Hankel): $p_3 \geq p_2^2$
- p_3 -OPPT ($n = 3$, optimal): $p_3 \geq \alpha x^3 + (1 - \alpha x)^3$
- p_n -PPT ($n = 3, 5, 7, \dots$): higher-order Hankel conditions
- p_n -OPPT ($n = 3, 5, 7, \dots$): higher-order optimal conditions

Detection Power

Fraction of entangled $D \times D$ states detected (Hilbert–Schmidt distribution)

D	NPT	NPT3	NPT5	ONPT3	ONPT4	NPT5
2	75.68%	25.53%	39.97%	75.68%	64.78%	75.68%
3	99.99%	25.32%	39.46%	91.63%	97.51%	98.97%
4	100%	23.29%	33.69%	98.68%	100%	100%
5	100%	21.80%	34.54%	99.95%	100%	100%
6	100%	20.93%	31.20%	100%	100%	100%

- **NPT**: PPT criterion (full spectrum)
- **NPT n** : p_n -PPT criterion
- **ONPT n** : p_n -OPPT criterion (optimal)

The optimal criterion ONPT3 detects **significantly more** states than NPT3, and approaches full NPT detection power. ONPT4 and NPT5 achieve near-complete detection.

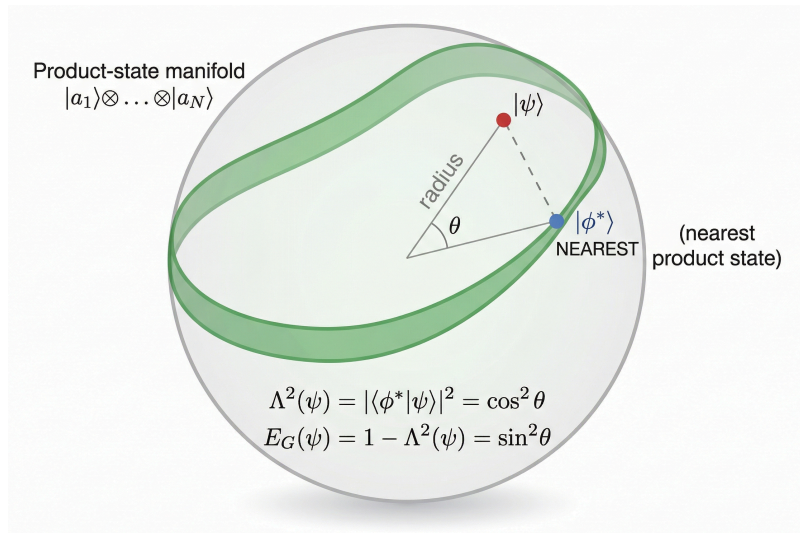
Summary: PT-Moment Certification

1. **Hankel matrix method (p_n -PPT):** relaxation to classical moment problems
 - Lowest order gives $p_3 \geq p_2^2$
 - Higher orders provide strictly stronger criteria
2. **Optimal method (p_n -OPPT):** exact solution of the PT-moment problem
 - Gives necessary and sufficient conditions
 - Detects significantly more entangled states
 - Dimension-independent
3. **Advantages over conventional methods**
 - **No full tomography required** — only PT-moments from randomized measurements
 - **No prior information** needed (unlike witness-based methods)
 - **Efficient measurement scaling** — higher-order moments cost only marginally more

Quantifying Entanglement from the Geometric Perspective

Weinbrenner and Gühne (EPL 2025)

Geometric Measure of Entanglement



The **geometric measure** quantifies entanglement by the distance to the nearest separable (product) state.

Definition: For a pure state $|\psi\rangle$ of N particles, define the **maximal overlap** with product states:

$$\Lambda^2(\psi) = \max_{|\phi\rangle = |a_1\rangle \otimes |a_2\rangle \otimes \dots \otimes |a_N\rangle} |\langle \phi | \psi \rangle|^2$$

The **geometric measure of entanglement** is $E_G(\psi) = 1 - \Lambda^2(\psi)$.

Schmidt Decomposition and Λ

Bipartite case

Any bipartite state can be written in the **Schmidt decomposition**:

$$|\psi\rangle = \sum_{i=1}^r s_i |a_i\rangle |b_i\rangle$$

where $s_1 \geq s_2 \geq \dots \geq s_r > 0$ are the **Schmidt coefficients**, $\sum_i s_i^2 = 1$, and r is the **Schmidt rank**.

The maximal overlap is simply the **largest Schmidt coefficient** $\Lambda(\psi) = s_1$.

The Schmidt decomposition follows from the SVD of the coefficient matrix τ_{ij} :

$$|\psi\rangle = \sum_{i,j} \tau_{ij} |i\rangle_A |j\rangle_B$$

A state is separable iff the Schmidt rank $r = 1$ (i.e., τ_{ij} has rank one and must be one).

Basic Examples

GHZ state

$$|\text{GHZ}_N\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N})$$

$$\Lambda^2 = \frac{1}{2}, \quad E_G = \frac{1}{2}$$

Nearest product state: $|0\rangle^{\otimes N}$ or $|1\rangle^{\otimes N}$, independent of N .

W state

$$|W_N\rangle = \frac{1}{\sqrt{N}}(|10\cdots 0\rangle + |01\cdots 0\rangle + \cdots + |0\cdots 01\rangle)$$

$$\Lambda^2 = \left(\frac{N-1}{N}\right)^{N-1} \xrightarrow{N \rightarrow \infty} \frac{1}{e}, \quad E_G = 1 - \Lambda^2$$

Computation and the Injective Tensor Norm

Multipartite states

For an N -partite state $|\psi\rangle = \sum_{i_1, \dots, i_N} T_{i_1 \dots i_N} |i_1 \dots i_N\rangle$, the maximal overlap is:

$$\Lambda(\psi) = \max_{|a_k\rangle} \left| \sum_{i_1, \dots, i_N} T_{i_1 \dots i_N} (a_1)_{i_1} \dots (a_N)_{i_N} \right|$$

Injective tensor norm

This is precisely the **injective tensor norm** of the coefficient tensor T :

$$\|T\|_\sigma = \max_{\|a_k\|=1} |T(a_1, a_2, \dots, a_N)| = \Lambda(\psi)$$

The injective tensor norm is also called the **spectral norm** of the tensor.

Remark: Computing $\Lambda(\psi)$ for multipartite states is in general **NP-hard**.

Why Tensor Eigenvalues?

The computational problem

Computing the geometric measure requires solving:

$$\Lambda(\psi) = \max_{\|a_k\|=1} \left| \sum_{i_1, \dots, i_N} T_{i_1 \dots i_N} (a_1)_{i_1} \cdots (a_N)_{i_N} \right|$$

This is a **polynomial optimization on the product of unit spheres** — a non-convex problem.

Why Tensor Eigenvalues?

Why eigenvalues?

For **matrices** ($N = 2$), the analogous problem $\max_{\|x\|=1} |x^T A y|$ is solved by the **singular value decomposition**. The critical points satisfy $Ay = \sigma x$, $A^T x = \sigma y$.

For **tensors** ($N \geq 3$), no SVD exists in general. Instead, the critical points of the optimization satisfy a **tensor eigenvalue equation**:

$$T\bar{x}^{N-1} = \lambda x, \quad \|x\| = 1$$

The **largest eigenvalue** gives $\Lambda(\psi)$. Tensor eigenvalue theory thus provides the natural mathematical framework for characterizing the extrema of the geometric measure.

Tensor Eigenvalues: Setup

The coefficient tensor $T_{i_1 i_2 \dots i_N}$ of a quantum state $|\psi\rangle$ defines a multilinear form. The geometric measure connects to the **eigenvalue theory of tensors**.

Tensor notation

For a tensor T of order N with dimensions $d_1 \times d_2 \times \dots \times d_N$, define the contraction:

$$T\bar{x}^{N-1} := \sum_{i_2, \dots, i_N} T_{i_1 i_2 \dots i_N} \bar{x}_{i_2}^{(2)} \dots \bar{x}_{i_N}^{(N)}$$

Eigenvalue equation

A scalar λ and vector x satisfy the **tensor eigenvalue equation** if:

$$T\bar{x}^{N-1} = \lambda x$$

with the normalization $x^T x = 1$ (or $x^\dagger x = 1$ in the complex case).

The **largest eigenvalue** of the tensor equals $\Lambda(\psi)$, the maximal overlap with product states.

Z-Eigenvalues

Definition (for real symmetric tensors)

A scalar $\lambda \in \mathbb{R}$ is a **Z-eigenvalue** of a real symmetric tensor T if there exists $x \in \mathbb{R}^d$ with $x^T x = 1$ such that:

$$T\bar{x}^{N-1} = \lambda x$$

Properties

- Z-eigenvalues always exist for symmetric tensors
- Every Z-eigenvalue is a **critical value** of the polynomial $T(x, x, \dots, x)$ on the unit sphere
- The **largest Z-eigenvalue** equals $\|T\|_\sigma$ (the injective tensor norm)

US-Eigenvalues

Beyond real tensors: unitary eigenvalues

For complex tensors (general quantum states), Z-eigenvalues are insufficient. The appropriate generalization uses **US-eigenvalues** (unitary-similarity eigenvalues).

Definition

A scalar λ is a **US-eigenvalue** of tensor T if there exist unitaries $U_2, \dots, U_N$ and a vector x with $x^\dagger x = 1$ such that:

$$T(I, U_2, \dots, U_N) \bar{x}^{N-1} = \lambda x$$

Key result

For a general N -partite state $|\psi\rangle$ with coefficient tensor T :

$$\Lambda(\psi) = \max\{|\lambda| : \lambda \text{ is a US-eigenvalue of } T\}$$

Asymptotic Behaviour of Generic States

Key result

For Haar-random N -qubit pure states ($N > 11$):

$$\text{Prob}(\Lambda^2(\psi) > 3N^2 \cdot 2^{-N}) < \exp(-N)$$

This means that for generic multi-qubit states, the maximal overlap with product states is **exponentially small**:

$$\Lambda^2(\psi) \lesssim 3N^2 \cdot 2^{-N}$$

Physical interpretation

- The geometric entanglement $E_G = 1 - \Lambda^2 \rightarrow 1$ for almost all states
- Generic states are highly entangled in the geometric sense

Mixed State Extension: Convex Roof

For mixed states ρ , the geometric measure might be extended via the convex roof construction:

$$\Lambda^2(\rho) = \max_{\{p_i, |\psi_i\rangle\}} \sum_i p_i \Lambda^2(|\psi_i\rangle) = \min_{\{p_i, |\psi_i\rangle\}} \sum_i p_i (1 - \Lambda^2(|\psi_i\rangle))$$

where the minimization is over all pure-state decompositions $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$.

The convex roof minimization is generally intractable. Analytical results exist only for special cases (e.g., permutation-symmetric states).

Summary: The Geometric Perspective

Concepts

- Injective tensor norm: $= \Lambda(\psi)$, maximal product state overlap
- Z-eigenvalues: Extrema on real unit sphere $\rightarrow \Lambda$ for real states
- US-eigenvalues: Complex generalization $\rightarrow \Lambda$ for arbitrary states

Open directions

- Extending the geometric measure to multipartite mixed states efficiently
- Connecting tensor rank to operational entanglement properties
- Better algorithms for computing the injective tensor norm

Thank You!

Questions & Discussion